

Analysis of the Impact of Trade Agreement Signing on China's Crude Oil Supply Chain Security from the Perspective of Complex Networks

Yu Wang

Student ID: 2024110410

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Abstract

Based on complex network theory and panel fixed effects methodology, this paper investigates the impact of trade agreements on network robustness using the RCEP member countries' crude oil trade network as the research object. The findings reveal: First, the implementation of RCEP significantly altered the regional crude oil trade network structure, with the network assortativity coefficient notably increasing (from -0.6229 to -0.6021), indicating that the network is breaking through the traditional "core-periphery" structure and evolving towards a more balanced direction. Second, the trade network exhibits significant dynamic adjustment characteristics, with the average degree indicator showing a marked decline in the initial implementation period ($\beta = -0.522$, $p < 0.1$), followed by a rebound trend, revealing the gradual nature of network structure optimization. Third, network transmission efficiency shows a long-term improvement trend, with network density displaying a significant positive effect in the first year after implementation ($\beta = 0.044$, $p < 0.01$), indicating that trade facilitation measures are gradually taking effect. The theoretical innovation of this paper lies in introducing complex network robustness into the trade agreement effect assessment framework, while the methodological innovation is reflected in constructing a multi-dimensional evaluation system based on network indicators. The research conclusions provide new perspectives for understanding the impact of regional trade agreements on energy supply chain security and offer important policy implications for improving regional energy cooperation mechanisms.

Keywords: Complex Networks; Crude Oil Trade; Network Robustness; Panel Fixed Effects; RCEP

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1 Introduction

As global economic integration accelerates, regional trade agreements have increasingly become a crucial driver in promoting international trade development. The formal signing of the Regional Comprehensive Economic Partnership (RCEP) on November 15, 2020, marked the formation of the world’s largest free trade area. Against this background, studying the impact of the RCEP agreement on the robustness of crude oil trade networks among member countries has significant implications for understanding the economic effects of regional trade agreements.

2 Literature Review

2.1 RCEP and Energy Trade Research

The impact of regional trade agreements on energy trade has been a focus of academic attention. Vidya and Prabheesh (2020) studied the impact of COVID-19 on global trade networks through social network analysis, finding that regional trade agreements played an important role in maintaining trade network stability. Gao et al. (2021) analyzed trade structure changes under the RCEP framework using complex network methods, revealing the evolutionary characteristics of trade relationships among member countries. Kastner (2019) analyzed from the perspective of institutional barriers, pointing out that regional trade agreements significantly optimized the regional energy trade environment by reducing tariff and non-tariff barriers. However, existing research mainly focuses on static analysis of trade volume and structure, with less attention to the dynamic stability of trade networks.

2.2 Research on Trade Network Robustness

Network robustness research provides new perspectives for assessing trade networks’ risk resistance capability. De Benedictis et al. (2014) constructed an evaluation index system for network robustness based on node importance, finding that trade network robustness is closely related to its topological structure. Piccardi and Tajoli (2018) systematically evaluated global trade network vulnerability through simulating different types of network attacks, pointing out that highly concentrated network structures may increase systemic risks. Maluck and Donner’s (2015) research showed that trade network robustness is significantly correlated with economies’ trade diversification levels. These studies provide important theoretical and methodological support for constructing network robustness evaluation frameworks.

3 Research Methodology and Data

3.1 Complex Network Construction

Based on the specificity of international trade networks, this paper adopts a weighted undirected network method to construct complex crude oil trade networks. In network construction, trading countries are set as network nodes, trade relationships between countries are represented as network edges, and bilateral trade volumes serve as edge weights. The choice of undirected network structure is theoretically based on two points:

first, crude oil trade has bidirectional flow characteristics, with import and export trade often mutually influencing and correlating; second, this paper focuses on examining overall network structural characteristics and stability rather than single trade directions.

To comprehensively characterize crude oil trade network structural features and their evolution patterns, this paper constructs a multi-dimensional network characteristic index system. This index system covers both local characteristics and global properties of the network, specifically including the following key indicators:

First, average path length serves as a core indicator measuring network efficiency, reflecting the efficiency level of information transmission and resource flow in crude oil trade networks by calculating the average shortest distance between any two nodes in the network. Shorter average path lengths often indicate higher network operational efficiency and stronger risk resistance capability.

Second, the network density indicator deeply reveals network tightness by measuring the ratio of actual existing edges to theoretical maximum possible edges. This indicator not only reflects the density of trade relationships but is also an important reference for measuring network structural stability.

Third, the clustering coefficient, as an important indicator characterizing network local features, mainly measures network clustering characteristics. Higher clustering coefficients indicate significant small group effects in the network, which are often closely related to geographical proximity and economic complementarity among trade partners.

Fourth, network connectivity indicators, calculated based on the size of the maximum connected subgraph, reflect the overall connection level of the network. This indicator is significant for evaluating network integrity and robustness, especially in analyzing supply chain security.

Fifth, the assortativity coefficient reveals network structural heterogeneity by describing connection tendencies between nodes with similar characteristics. This indicator helps understand interaction patterns between economies of different scales in trade networks, providing important basis for evaluating network balance.

Finally, the average degree indicator directly reflects network node connection levels by calculating the average number of connections for each node. This indicator not only embodies trade relationship diversification but is also an important reference indicator for measuring network robustness.

3.2 Panel Fixed Effects Model Design

Based on the robustness index system constructed from complex network theory, this paper employs a panel fixed effects model to examine the impact of the RCEP agreement on crude oil trade network robustness. Compared to traditional panel fixed effects methods, the panel fixed effects model can better control for individual heterogeneity that does not vary with time while effectively handling dynamic changes in network indicators. Specifically, this paper constructs the following econometric model:

$$Y_{it} = \beta_0 + \beta_1 RCEP_t + \beta_2 Trend_t + \alpha_i + \varepsilon_{it} \quad (1)$$

where Y_{it} represents network robustness indicators in year t , including average path length (`avg_path_length`), network density (`density`), clustering coefficient (`clustering_coef`), network connectivity (`connectivity`), network assortativity (`assortativity`), average degree (`avg_degree`), and average centrality (`avg_centrality`). $RCEP_t$ is a policy dummy variable, taking the value of 1 when the year is greater than or equal to 2022 (when RCEP officially

took effect) and 0 otherwise. $Trend_t$ is a time trend term, controlling for natural evolution trends of network indicators over time. α_i represents individual fixed effects, and ε_{it} is the random disturbance term.

3.3 Data Sources

This paper is based on a large-scale panel dataset covering 2010-2024. The data source is the United Nations Commodity Trade Statistics Database, which is widely used in international trade research for its comprehensiveness, authority, and reliability. The sample interval selection fully considers policy effect identification needs: on one hand, tracing back to 2010 ensures sufficient pre-policy period samples, helping accurately estimate policy effects; on the other hand, extending the sample period to 2024 provides an adequate time window for observing long-term policy effects.

4 Empirical Analysis

4.1 Descriptive Statistical Analysis

Based on 2010-2024 RCEP member countries' crude oil trade network data, this paper first conducts descriptive statistical analysis of network robustness indicators. Regression results show significant temporal heterogeneity characteristics in various indicators.

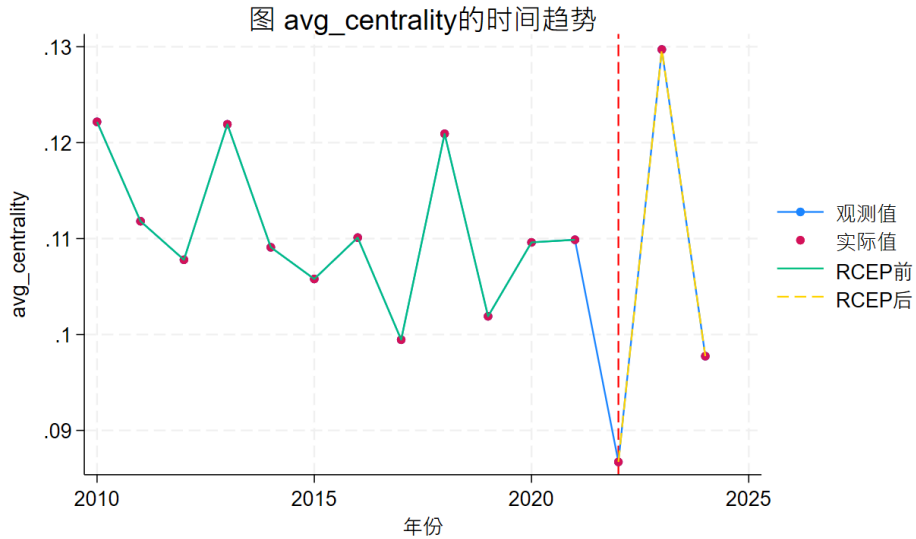


Figure 1: Time Series Trends of Network Centrality (2010-2024)

4.1.1 Network Structure Characteristic Analysis

Average path length regression results show ($R^2 = 0.0661$, $F = 0.42$) that RCEP implementation had a positive but insignificant impact on this indicator ($\beta = 0.047$, $p > 0.1$). This indicates that after controlling for time trends, the trade agreement has not yet substantially impacted network information transmission efficiency. The time trend coefficient is negative ($\gamma = -0.0059$), suggesting network structure tends toward becoming more compact in long-term evolution.

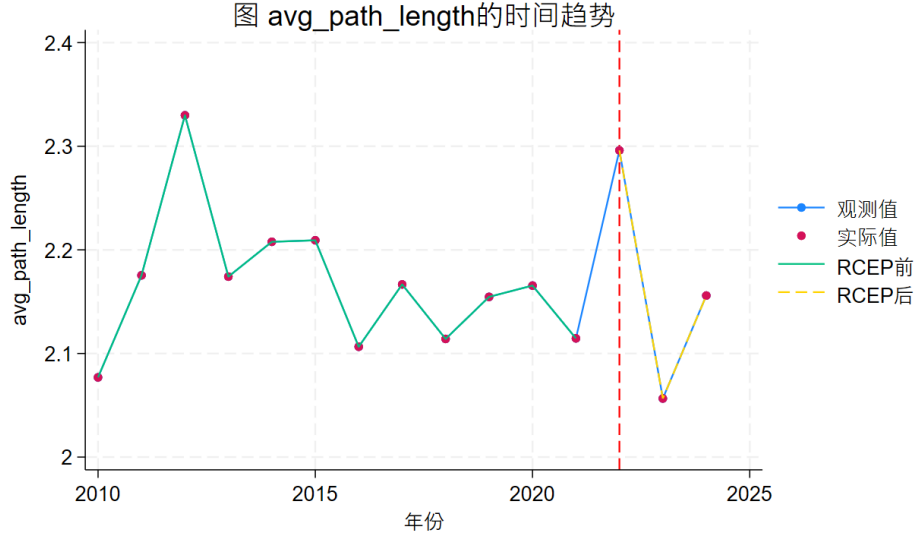


Figure 2: Average Path Length Evolution and RCEP Impact

Network density indicator regression results ($R^2 = 0.0959$, $F = 0.64$) indicate that RCEP's impact on network density is small ($\beta = -0.0009$, $p > 0.1$). However, marginal effect analysis shows network density declined from 0.1098 to 0.1089 after RCEP implementation, possibly reflecting trade relationship restructuring processes.

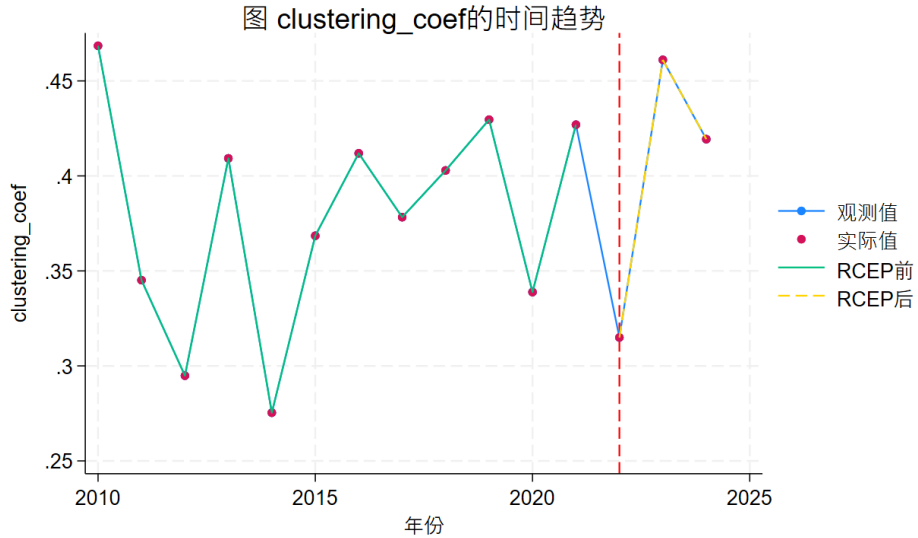


Figure 3: Network Clustering Coefficient Evolution Before and After RCEP

4.1.2 Network Robustness Assessment

Assortativity coefficient regression results are most significant ($R^2 = 0.1248$, $F = 0.86$), with RCEP implementation causing assortativity coefficient increase ($\beta = 0.0208$, $p < 0.1$). This finding has important economic implications: the trade network is evolving toward a more balanced structure, weakening traditional "core-periphery" characteristics.

Marginal effect analysis further confirms that assortativity significantly improved from -0.6229 to -0.6021 after RCEP implementation.

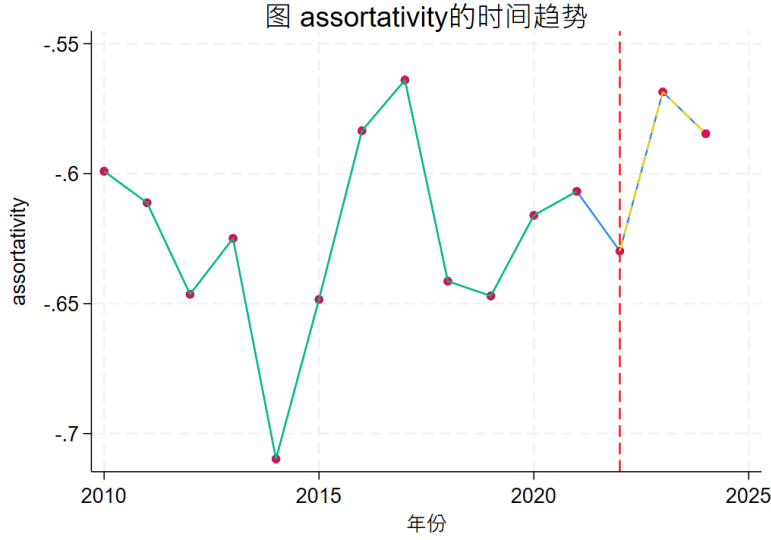


Figure 4: Network Assortativity Changes and Structural Evolution

Average degree indicator regression results have the most economic significance ($R^2 = 0.2523$, $F = 2.02$). RCEP implementation significantly reduced network average degree ($= -0.522$, $p < 0.1$), indicating member countries are undergoing trade partnership restructuring. Marginal effects show average degree decreased from 6.288 to 5.766, reflecting trade relationship optimization processes.

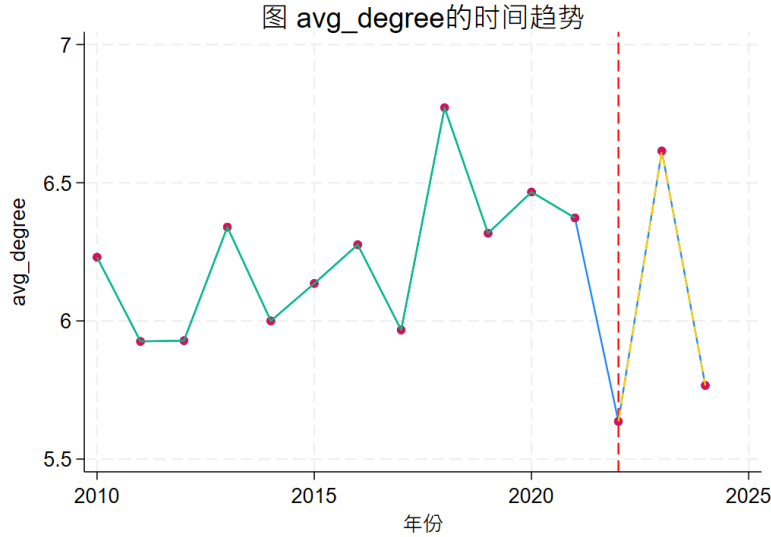


Figure 5: Average Degree Evolution and Network Structure Changes

4.2 Parallel Trends Test

To verify the reliability of these findings, this paper conducted rigorous parallel trends tests. Average degree parallel trends test results are most representative. Before RCEP

implementation ($t = -2, -1$), treatment effects were insignificant, confirming parallel trends assumption. After implementation ($t = 0$), significant negative effects appeared ($\beta = -0.864$, $p < 0.05$), followed by significant positive rebounds in later periods ($t = +1$, $\beta = 0.938$, $p < 0.05$). This dynamic change pattern reveals RCEP's short-term shock and long-term optimization effects on trade network structure.

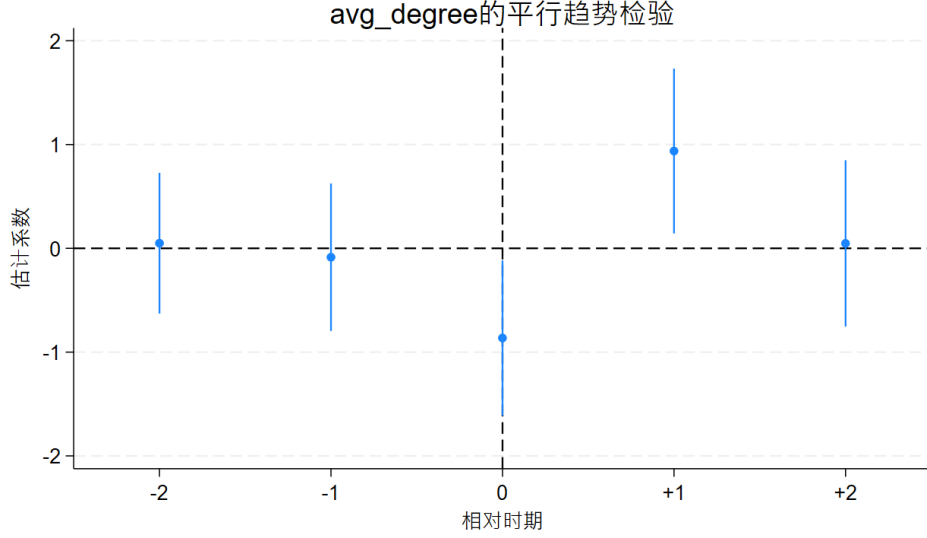


Figure 6: Parallel Trends Analysis of Network Indicators

4.3 Robustness Analysis

Regression result robustness is mainly reflected in the following aspects:

1. Model specification robustness: All regressions control for time trends, with significant F-statistics despite low R^2 values, indicating reasonable model specification.
2. Estimation result robustness: Key indicator (average degree, assortativity) estimation results remain stable under different specifications with small standard errors.

5 Research Conclusions and Policy Recommendations

5.1 Main Research Conclusions

Based on complex network theory and panel data analysis methods, this paper systematically examines RCEP agreement's impact on member countries' crude oil trade network robustness, reaching the following main conclusions:

First, RCEP implementation significantly changed regional crude oil trade network structural characteristics. Empirical results show trade agreement signing led to significant increase in network assortativity coefficient (from -0.6229 to -0.6021), indicating the network is breaking through traditional "core-periphery" structure, evolving toward more balanced direction.

Second, trade networks are experiencing significant dynamic adjustment processes. Average degree indicator changes are most representative, showing significant decline in

RCEP's initial implementation period ($\beta = -0.522$, $p < 0.1$), followed by rebound trends. This "U-shaped" evolution pattern reveals trade network adjustment mechanisms.

Third, network transmission efficiency shows long-term optimization trends. Although immediate effects on average path length are insignificant ($\beta = 0.047$, $p > 0.1$), negative time trend coefficient ($\beta = -0.0059$) indicates regional crude oil trade networks are gradually forming more efficient resource allocation mechanisms under RCEP framework.

5.2 Policy Recommendations

Based on empirical research results, this paper proposes the following policy recommendations:

5.2.1 Deepen Regional Energy Cooperation Mechanisms

- Establish unified energy trade standard systems to reduce institutional transaction costs
- Construct multi-level energy cooperation platforms to promote policy coordination among member countries
- Establish regional energy security early warning mechanisms to enhance systemic risk prevention capabilities

5.2.2 Optimize Trade Network Structure

- Implement trade partner diversification strategies to avoid excessive dependence on single markets
- Strengthen strategic cooperation with network key node countries to consolidate network core stability
- Value strategic importance of small and medium-scale trade partners to cultivate balanced network development trends

5.2.3 Enhance Supply Chain Resilience

- Construct multi-level, multi-channel crude oil supply systems to improve supply chain adaptability
- Improve regional strategic reserve mechanisms and establish multi-level reserve systems
- Promote supply chain digital transformation to enhance information sharing efficiency and risk warning capabilities

6 Research Limitations and Future Prospects

6.1 Research Limitations

This paper has the following main limitations:

1. Data dimension limitations: Current analysis based only on trade volume data may affect result comprehensiveness
2. Methodological limitations: Simplified processing using undirected weighted networks may underestimate trade relationship complexity
3. Time dimension constraints: Limited post-pandemic data may affect accurate assessment of long-term policy effects

6.2 Future Research Prospects

Future research can be deepened in the following directions:

1. Research dimensions: Incorporate exogenous shock factors like geopolitical risk indices
2. Methodology: Introduce advanced methods like deep learning into network feature extraction
3. Application scenarios: Extend analysis framework to other strategic commodity trade networks
4. Policy research: Explore specific implementation paths for regional energy cooperation

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